PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED
(TOTAL DOSE AND SINGLE EVENT EFFECTS) TRANSISTORS, N-CHANNEL
SILICON TYPES 2N7395, 2N7396, 2N7397, AND 2N7398
JANTXD , -R, JANTXVD , -R, AND JANSD, -R

This specification is approved for use by all Departments and
Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a N-Channel, enhancement-mode, MOSFET, radiation hardened (Total Dose and Single Event characterization - see figure 4), power transistor intended for use in high density power switching applications. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1, T0-205AF.

1.3 Maximum ratings. $T_A$ = +25°C, unless otherwise specified.

| Type | $P_T$ 1/ $T_C$ =+25°C | $P_T$ 1/ $T_A$ =+25°C | $V_{DS}$ | $V_{DG}$ | $V_{GS}$ | $I_{D1}$ 2/ $T_C$ =+25°C | $I_{D2}$ $T_C$=+100°C | $I_D$ 2/ | $I_{DM}$ | $T_J$ and $T_{STG}$ | $V_{ISO}$ 70,000 ft. altitude |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | W | W | V dc | V dc | V dc | A dc | A dc | A dc | A (pk) | °C | V dc |
| 2N7395 | 25 | 10 | 100 | 100 | ±20 | 8.0 | 5.0 | 8.0 | 24 | -55 | N/A |
| 2N7396 | | | 200 | 200 | | 5.0 | 3.0 | 5.0 | 15 | to | N/A |
| 2N7397 | | | 250 | 250 | | 4.0 | 2.0 | 4.0 | 12 | +150 | 250 |
| 2N7398 | | | 500 | 500 | | 2.0 | 1.0 | 2.0 | 6 | | 500 |

1/ Derate linearly 0.2 W/°C for $T_C$ > +25°C; $P_T = (T_{jmax} - T_C)/R_{\Theta JC}$.
2/ $I_D = ((T_{jmax} - T_C)/((R_{\Theta JC}) \times (r_{DS(on)} \text{ at } T_{jmax})))^{1/2}$

1.4 Primary electrical characteristics at $T_c$ = +25°C.

| Type | Min $V_{(BR)DSS}$ $V_{GSS}$ = 0 $I_D$ = 1.0 mA dc | $V_{GS(TH)1}$ $V_{DS} \geq V_{GS}$ $I_D$ = 1.0 mA dc | | Max $I_{DSS1}$ $V_{GS}$ = 0 $V_{GS}$ = 80% of rated $V_{DS}$ | Max $r_{DS(on)}$ 1/ $V_{GS}$ = 12V | | $R_{\Theta JC}$ Max | $I_{AS} = I_{DM}$ |
|---|---|---|---|---|---|---|---|---|
| | | | | | $T_J$ = 25°C at $I_{D2}$ | $T_J$ = 125°C at $I_{D2}$ | | |
| | V dc | V dc Min | V dc Max | µA dc | Ω | Ω | °C/W | A (pk) |
| 2N7395 | 100 | 1.5 | 4.0 | 25 | 0.23 | 0.36 | 5.00 | 24 |
| 2N7396 | 200 | | | | 0.46 | 0.81 | | 15 |
| 2N7397 | 250 | | | | 0.61 | 1.11 | | 12 |
| 2N7398 | 500 | | | | 2.50 | 4.80 | | 6 |

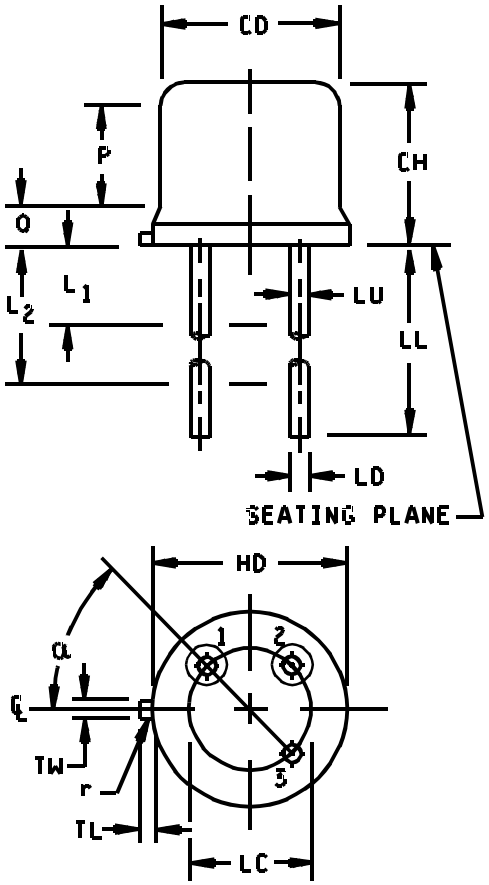
1/ Pulsed (see 4.5.1).

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of *
* use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, *
* ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization *
* Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. *


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| Symbol | Millimeters | | Inches | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| CD | 8.01 | 8.50 | .315 | .335 |
| CH | 4.07 | 4.57 | .160 | .180 |
| HD | 8.89 | 9.39 | .350 | .370 |
| LC | 2.42 | 2.66 | .095 | .105 |
| LD | 0.41 | 0.53 | .016 | .021 |
| LL | 12.70 | 14.22 | .500 | .560 |
| P | 1.91 | --- | .075 | --- |
| TL | .740 | 1.14 | .029 | .045 |
| TW | .720 | .860 | .028 | .034 |
| r | --- | .025 | --- | .010 |
| α | 45° TP | | 45° TP | |
| Term 1 | Drain | | | |
| Term 2 | Source | | | |
| Term 3 | Gate | | | |




NOTES:
1. Dimension are in milimeters
2. Equivalents are given for general information only.
3. Terminals 1 and 2 are isolated from the case, terminal 3 is butt welded to stem base.
4. The preferred measurements used herein are the metric units. However, this transistor was designed using inch -pound units of measurement. In case of conflicts between the metric and inch-pound units, the inch-pound units shall rule.
5. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions (similar to TO-205AF).



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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figure 1 herein.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52; a copper core or plated core is permitted. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor but shall be retained on the initial container.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic discharge protection (see 6.2).



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3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended (see 3.5).

a. Devices should be handled on benches with conductive handling devices.
b. Ground test equipment, tools and personnel handling devices.
c. Do not handle devices by the leads.
d. Store devices in conductive foam or carriers.
e. Avoid use of plastic, rubber or silk in MOS areas.
f. Maintain relative humidity above 50 percent if practical.
g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.
h. Gate must be terminated to source, $R \leq$ or 100 kΩ, whenever bias voltage is applied drain to source.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in paragraphs 4.4.2 and 4.4.3.

3.8 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for qualification inspection in accordance with figure 4 of MIL-PRF-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500, and table III herein. End point electrical measurements shall be in accordance with the applicable steps of table IV herein.



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4.3 Screening (JANS, JANTX, and JANTXV). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table IV of MIL-PRF-19500) | Measurement | |
|---|---|---|
| | JANS | JANTX and JANTXV |
| 1/ | $E_{AS}$ test (see 4.5.4) | $E_{AS}$ test (see 4.5.4) |
| 1/ | Method 3161 (see 4.5.3) | Method 3161 (see 4.5.3) |
| 1/ | Gate stress test (see 4.5.5) | Gate stress test (see 4.5.5) |
| 2/ | Subgroup 2 of table I herein | Subgroup 2 of table I herein |
| 9 | $I_{GSS}$, $I_{DSS1}$ as a minimum | Not applicable |
| 10 | MIL-STD-750, method 1042 test condition B | MIL-STD-750, method 1042 test condition B |
| 11 | $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, $r_{DS(ON)}$, $V_{GS(TH)}$ Subgroup 2 of table I herein.<br><br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100% of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100% of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±10 µA dc or ±100% of initial value, whichever is greater. | $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, $r_{DS(ON)}$, $V_{GS(TH)}$ Subgroup 2 of table I herein. |
| 12 | MIL-STD-750, method 1042 test condition A | MIL-STD-750, method 1042 test condition A |
| 13 | Subgroups 2 and 3 of table I herein<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100% of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100% of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±10 µA dc or ±100% of initial value, whichever is greater.<br>$\Delta r_{DS(ON)1}$ = ±20% of initial value.<br>$\Delta V_{GS(TH)1}$ = ±20% of initial value. | Subgroups 2 and 3 of table I herein<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100% of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100% of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±10 µA dc or ±100% of initial value, whichever is greater.<br>$\Delta r_{DS(ON)1}$ = ±20% of initial value.<br>$\Delta V_{GS(TH)1}$ = ±20% of initial value. |

1/ Shall be performed anytime before screen 10.
2/ Shall be performed after $E_{AS}$ test, method 3161, and gate stress test.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for conformance inspection in accordance with figure 4 of MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table V of MIL-PRF-19500 and table I herein.



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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JANTX and JANTXV) of MIL-PRF-19500 and as follows. End point electrical measurements shall be in accordance with the applicable steps of table IV herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| 3 | 1051 | Temp. cycling, test condition G, 100 cycles |
| 3 | 2077 | SEM |
| 4 | 1042 | Intermittent operation life, condition D, 2,000 cycles. No heat sink nor or forced-air cooling on the device shall be permitted during the on cycle. $t_{ON}$ = 30 seconds minimum. |
| 5 | 1042 | Accelerated steady-state reverse bias, condition A, $V_{DS}$ = rated; $T_A$ = +175°C; t = 120 hours, minimum. |
| 5 | 1042 | Accelerated steady-state gate bias, condition B, $V_{GS}$ = rated; $T_A$ = +175°C; t = 24 hours. |
| 5 | 2037 | Bond strength (Al-Au die interconnects only), Test condition A |
| 6 | 3161 | Thermal resistance, see 4.5.2. |

4.4.2.2 Group B inspection, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| 2 | 1051 | Temp. cycling, test condition G, 25 cycles. |
| 3 | 1042 | Intermittent operation life (LTPD), test condition D, 2,000 cycles, 1 cycle = 30 s min, |
| 5 and 6 | ---- | Not applicable. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500 and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

| Subgroup | Method | Condition |
|---|---|---|
| 2 | 2036 | Terminal strength, test condition E, weight = 8 oz., 3 arcs. |
| 6 | 1042 | Intermittent operation life, test condition D, 6,000 cycles; 1 cycle = 30 sec. min. |

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with table VIII of MIL-PRF-19500 and table IV herein.

4.4.4.1 Design parameters. Not tested on a per lot basis. Design shall be such that the devices shall be capable of meeting the requirements in figure 4.



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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal Resistance. Thermal Resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit of $R_{\theta JC}$ = 5.0 °C/W. The following parameters shall apply:

a. Measuring current($I_M$)..... 10 mA

b. Drain heating current ($I_H$)..... 1 A

c. Heating time ($t_H$).....Steady-state (see MIL-STD-750, method 3161)

d. Drain-source heating voltage ($V_H$)..... 25 V

e. Measurement time delay ($t_{MD}$)..... 30 to 60 ms

f. Sample window time ($t_{SW}$)..... 10 ms maximum

4.5.3 Thermal Response ($\Delta V_{SD}$ measurement). The delta $V_{SD}$ measurement shall be performed in accordance with method 3161 of MIL-STD-750. The delta $V_{SD}$ conditions ($I_H$ and $V_H$) and maximum limit shall be derived by each vendor from the thermal response curves (see figure 2) and shall be specified in the certificate of conformance prior to qualification. The following parameter measurements shall apply:

a. Measuring current ($I_M$)..... 10 mA

b. Drain heating current ($I_H$)..... 2 A

c. Heating time ($t_H$)..... 10 ms

d. Drain-source heating voltage ($V_H$)..... 25 V

e. Measurement time delay ($t_{MD}$)..... 30 - 60 µs

f. Sample window time ($t_{SW}$)..... 10 µs maximum

4.5.4 Single pulse avalanche energy ($E_{AS}$).

a. $I_{AS} = I_{DM}$

b. L = 0.1 mH

c. $E_{AS} = 1/2\ LI_{AS}^2$

d. $V_{DD}$ = 50 V to 150 V dc

e. Initial junction temperature = 25 °C, -5 °C, +10 °C

4.5.5 Gate stress test.

a. $V_{GS}$ = 30 V

b. t = 250 µS, minimum



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TABLE I. Group A inspection

| Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical inspection | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Breakdown voltage drain to source | 3407 | $V_{GS}$ = 0V, $I_D$ = 1 mA dc, Bias condition C | $V_{(BR)DSS}$ | | | |
| 2N7395 | | | | 100 | | V dc |
| 2N7396 | | | | 200 | | V dc |
| 2N7397 | | | | 250 | | V dc |
| 2N7398 | | | | 500 | | V dc |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1 mA dc | $V_{GS(th)}$ | 1.5 | 4.0 | V dc |
| Gate current | 3411 | $V_{GS}$ = +20V dc and -20V dc, Bias condition C, $V_{DS}$ = 0V | $I_{GSS1}$ | | ±100 | nA dc |
| Drain current | 3413 | $V_{GS}$ = 0V dc, Bias condition C, $V_{DS}$ = 80 % of rated $V_{DS}$, | $I_{DSS1}$ | | 25 | mA dc |
| Static drain to source "ON" state resistance | 3421 | $V_{GS}$ = 12V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D2}$ | $r_{DS(on)1}$ | | | |
| 2N7395 | | | | | 0.23 | Ω |
| 2N7396 | | | | | 0.46 | Ω |
| 2N7397 | | | | | 0.61 | Ω |
| 2N7398 | | | | | 2.50 | Ω |
| Static drain to source "ON" state voltage | 3405 | $V_{GS}$ = 12V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D1}$ | $V_{DS(on)}$ | | | |
| 2N7395 | | | | | 1.93 | V dc |
| 2N7396 | | | | | 2.42 | V dc |
| 2N7397 | | | | | 2.56 | V dc |
| 2N7398 | | | | | 5.25 | V dc |
| Forward voltage | 4011 | $V_{GS}$ = 0 V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D1}$ | $V_{SD}$ | | 1.8 | V dc |
| Subgroup 3 | | | | | | |
| High temperature operation | | $T_C = T_J$ = +125°C | | | | |
| Gate current | 3411 | $V_{GS}$ = +20V dc and -20V dc, Bias condition C, $V_{DS}$ = 0V | $I_{GSS2}$ | | ±200 | nA dc |
| Drain current | 3413 | $V_{GS}$ = 0V dc, Bias condition C, $V_{DS}$ = 80% of rated $V_{DS}$ | $I_{DSS2}$ | | 0.25 | mA dc |

See footnotes at end of table.



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TABLE I. Group A inspection

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 3-continued | | | | | | |
| Static drain to source "ON"-state resistance | 3421 | $V_{GS}$ = 12V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D2}$ | $r_{DS(on)2}$ | | | |
| 2N7395 | | | | | 0.36 | Ω |
| 2N7396 | | | | | 0.81 | Ω |
| 2N7397 | | | | | 1.11 | Ω |
| 2N7398 | | | | | 4.80 | Ω |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1 mA dc | $V_{GS(th)2}$ | 0.5 | | V dc |
| Low temperature operation | | $T_C = T_J$ = -55°C | | | | |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$ $I_D$ = 1 mA dc | $V_{GS(th)3}$ | | 5.0 | V dc |
| Subgroup 4 | | | | | | |
| Switching time test | 3472 | $I_D = I_{D1}$, $V_{GS}$ = 12 V dc<br>$R_G$ = 7.5 Ω, $V_{DD}$ = 50% of rated $V_{DS}$ | | | | |
| Turn-on delay time | | | $t_{D(on)}$ | | | |
| 2N7395 | | | | | 70 | ns |
| 2N7396 | | | | | 65 | ns |
| 2N7397 | | | | | 60 | ns |
| 2N7398 | | | | | 80 | ns |
| Rise Time | | | $t_R$ | | | |
| 2N7395 | | | | | 220 | ns |
| 2N7396 | | | | | 150 | ns |
| 2N7397 | | | | | 140 | ns |
| 2N7398 | | | | | 100 | ns |
| Turn-off delay time | | | $t_{D(off)}$ | | | |
| 2N7395 | | | | | 100 | ns |
| 2N7396 | | | | | 120 | ns |
| 2N7397 | | | | | 120 | ns |
| 2N7398 | | | | | 150 | ns |
| Fall time | | | $t_f$ | | | |
| 2N7395 | | | | | 90 | ns |
| 2N7396 | | | | | 85 | ns |
| 2N7397 | | | | | 95 | ns |
| 2N7398 | | | | | 140 | ns |

See footnotes at end of table.



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TABLE I. Group A inspection-continued

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 5 | | | | | | |
| Safe operating area test (high voltage) | 3474 | See figure 3, $t_p$ = 10 ms min. $V_{DS}$ = 80% of max. rated $V_{DS}$ ($V_{DS} \leq 200$ V) | | | | |
| Electrical measurerments | | See table IV, steps 1,2,3,4,5, 6, and 7. | | | | |
| Subgroup 6 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 7 | | | | | | |
| Gate charge | | | | | | |
| On-state gate charge | 3471 | Condition B | $Q_{G(on)}$ | | | |
| 2N7395 | | | | | | |
| 2N7396 | | | | | 43 | nC |
| 2N7397 | | | | | 38 | nC |
| 2N7398 | | | | | 40 | nC |
| | | | | | 35 | nC |
| Gate to source charge | | | | | | |
| 2N7395 | | | $Q_{GS}$ | | | |
| 2N7396 | | | | | 8.7 | nC |
| 2N7397 | | | | | 8.4 | nC |
| 2N7398 | | | | | 8.4 | nC |
| | | | | | 6.7 | nC |
| Gate to drain charge | | | | | | |
| 2N7395 | | | $Q_{GD}$ | | | |
| 2N7396 | | | | | 22 | nC |
| 2N7397 | | | | | 20 | nC |
| 2N7398 | | | | | 20 | nC |
| | | | | | 16 | nC |
| Reverse recovery time | | | | | | |
| 2N7395 | 3473 | di/dt = 100 A/µs, | $t_{RR}$ | | | |
| 2N7396 | | $V_{DD} \leq 30$ V | | | 330 | ns |
| 2N7397 | | $I_D = I_{D1}$ | | | 360 | ns |
| 2N7398 | | | | | 400 | ns |
| | | | | | 390 | ns |

1/ For sampling plan, see MIL-PRF-19500.



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Table II. Group D Inspection

| Inspection 1/ 2/ 3/ 4/ | MIL-STD-750 | | Symbol | Pre-irradiation limits | | Post irradiation limits | | Units |
|---|---|---|---|---|---|---|---|---|
| | Method | Conditions | | Min. | Max. | Min. | Max. | |
| Subgroup 1 | | | | | | | | |
| Not Applicable | | | | | | | | |
| Subgroup 2 | | $T_C$ = +25°C | | | | | | |
| Steady state total dose irradiation ($V_{GS}$ bias) 4/ | 1019 | $V_{GS}$ = 12V, $V_{DS}$ = 0V | | | | | | |
| Steady state total dose irradiation ($V_{DS}$ bias) 5/ | 1019 | $V_{GS}$ = 0V, $V_{DS}$ = 80% of rated $V_{DS}$ | | | | | | |
| Breakdown voltage drain to source | 3407 | $V_{GS}$ = 0V, $I_D$ = 1 mA dc, Bias condition C | $V_{(BR)DSS}$ | | | | | |
| 2N7395 | | | | 100 | | 100 | | V dc |
| 2N7396 | | | | 200 | | 200 | | V dc |
| 2N7397 | | | | 250 | | 250 | | V dc |
| 2N7398 | | | | 500 | | 500 | | V dc |
| Gate to source Voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1 mA dc | $V_{GS(TH)1}$ | 1.5 | 4.0 | 1.5 | 4.0 | V dc |
| Gate current | 3411 | $V_{GS}$ = +20V dc and -20 V dc, $V_{DS}$ = 0V,Bias condition C | $I_{GSS1}$ | | ±100 | | ±100 | nA dc |
| Drain current | 3413 | $V_{GS}$ = 0V, $V_{DS}$ = 80% of rated $V_{DS}$, Bias condition C | $I_{DSS1}$ | | 25 | | 25 | μA dc |
| Static drain to source "ON"-state resistance | 3421 | $V_{GS}$ = 12V dc, condition A, pulsed (see4.5.1), $I_D = I_{D2}$ | $r_{DS(ON)1}$ | | | | | |
| 2N7395 | | | | | 0.23 | | 0.23 | Ω |
| 2N7396 | | | | | 0.46 | | 0.46 | Ω |
| 2N7397 | | | | | 0.61 | | 0.61 | Ω |
| 2N7398 | | | | | 2.50 | | 2.50 | Ω |
| Static drain to source "ON"-state voltage | 3405 | $V_{GS}$ = 12V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D1}$ | $V_{DS(ON)}$ | | | | | |
| 2N7395 | | | | | 1.93 | | 1.93 | V dc |
| 2N7396 | | | | | 2.42 | | 2.42 | V dc |
| 2N7397 | | | | | 2.56 | | 2.56 | V dc |
| 2N7398 | | | | | 5.25 | | 5.25 | V dc |

1/ For sampling plan see MIL-PRF-19500
2/ Electrical specifications are for 'D' and 'R' rad levels.
3/ Group D qualification may be performed prior to lot formation. Wafers qualified to these group D QCI requirements may be used for any other detail specification utilizing the same die design.
4/ At the manufacturer's option, group D samples need not be subjected to all the screening tests, but shall be assembled in its qualified package or in any qualified package that the manufacturer has data to correlate the performance to the designated package.



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TABLE III. Group E inspection (all quality levels) - For qualification only

| Inspection 1/, 2/, 3/, 4/, 5/, 6/ | MIL-STD-750 | | Qualification and large lot quality conformance inspection. |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 12 devices<br>c = 0 |
| Temperature cycling (air to air) | 1051 | Test condition G, 500 cycles | |
| Hermetic seal | 1071 | | |
| Fine leak<br>Gross leak | | | |
| Electrical measurements | | See table IV, steps 1,2,3,4,5, 6, and 7 | |
| Subgroup 2 1/ | | | 12 devices<br>c = 0 |
| Steady-state gate bias | 1042 | Test condition B; 1,000 hours | |
| Electrical measurements | | See table IV, steps 1,2,3,4,5, 6, and 7 | |
| Steady state reverse bias | 1042 | Test condition A; 1,000 hours | |
| Electrical measurements | | See table IV, steps 1,2,3,4,5, 6, and 7. | |
| Subgroup 3 | | | |
| Not applicable | | | |
| Subgroup 4 | | | 22 devices<br>c = 0 |
| Thermal resistance | 3161 | $R_{\theta JC}$ = 5.0 °C/W maximum. See 4.5.2 | |
| Subgroup 5 | | | 15 devices<br>c = 0 |
| Barometric pressure test (not required for $V_{BR(DSS)} \leq 200V$)<br>2N7397<br>2N7398 | 1001 | Test condition C<br><br>$V_{DS}$ = 250 V; $I_{(ISO)}$ < 0.25 mA<br>$V_{DS}$ = 500 V; $I_{(ISO)}$ < 0.25 mA | |

1/ See footnotes at end of table.



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TABLE III. Group E inspection (all quality levels) - For qualification only - continued.

| Inspection 1/, 2/, 3/, 4/, 5/, 6/ | MIL-STD-750 | | Qualification and large lot conformance inspection. |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 6 | | | 3 devices 6/ c = 0 |
| Electrical measurements 4/ | | See table IV, steps 3 and 4 | |
| SEE Effect Testing 5/ | | Fluence = 3e5 ± 20% ions/cm$^2$<br>Flux = 5e3 to 2e4 ions/cm$^2$sec<br>Beam energy = 260 to 300 MeV<br>Temperature = 25°C ± 5°C | |
| 2N7395 | | LET = 26 to 30 MeV-cm$^2$/mg<br>Ion range = 40 to 45 microns<br>Insitu bias conditions: $V_{DS}$ = 100V and $V_{GS}$ = -20V<br><br>LET = 36 to 40 MeV-cm$^2$/mg<br>Ion range = 35 to 40 microns<br>Insitu bias conditions: $V_{DS}$ = 100V and $V_{GS}$ = -10V<br>$V_{DS}$ = 80V and $V_{GS}$ = -15V<br>$V_{DS}$ = 50V and $V_{GS}$ = -20V | |
| 2N7396 | | LET = 26 to 30 MeV-cm$^2$/mg<br>Ion range = 40 to 45 microns<br>Insitu bias conditions: $V_{DS}$ = 200V and $V_{GS}$ = -20V<br><br>LET = 36 to 40 MeV-cm$^2$/mg<br>Ion range = 35 to 40 microns<br>Insitu bias conditions: $V_{DS}$ = 200V and $V_{GS}$ = -5V<br>$V_{DS}$ = 160V and $V_{GS}$ = -10V<br>$V_{DS}$ = 100V and $V_{GS}$ = -15V<br>$V_{DS}$ = 40V and $V_{GS}$ = -20V | |
| 2N7397 | | LET = 26 to 30 MeV-cm$^2$/mg<br>Ion range = 40 to 45 microns<br>Insitu bias conditions: $V_{DS}$ = 250V and $V_{GS}$ = -20V<br><br>LET = 36 to 40 MeV-cm$^2$/mg<br>Ion range = 35 to 40 microns<br>Insitu bias conditions: $V_{DS}$ = 250V and $V_{GS}$ = -5V<br>$V_{DS}$ = 200V and $V_{GS}$ = -10V<br>$V_{DS}$ = 125V and $V_{GS}$ = -15V<br>$V_{DS}$ = 50V and $V_{GS}$ = -20V | |
| 2N7398 | | LET = 26 to 30 MeV-cm$^2$/mg<br>Ion range = 40 to 45 microns<br>Insitu bias conditions: $V_{DS}$ = 500V and $V_{GS}$ = -15V<br>$V_{DS}$ = 450V and $V_{GS}$ = -20V<br><br>LET = 36 to 40 MeV-cm$^2$/mg<br>Ion range = 35 to 40 microns<br>Insitu bias conditions: $V_{DS}$ = 500V and $V_{GS}$ = -5V<br>$V_{DS}$ = 400V and $V_{GS}$ = -10V<br>$V_{DS}$ = 100V and $V_{GS}$ = -15V | |

See footnote at end of table.



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TABLE III. Group E inspection (all quality levels) - For qualification only - continued.

| Inspection 1/, 2/, 3/, 4/, 5/, 6/ | MIL-STD-750 | | Qualification and large lot conformance inspection. |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 6<br><br>Electrical measurements 4/ | | See table IV, steps 3 and 4 | |

1/ A separate sample for each test may be pulled.
2/ Group E qualification of single event effect testing may be performed prior to lot formation. Wafers qualified to these group E QCI requirements may be used for any other performance specification utilizing the same die design.
3/ See Appendix A for appropriate method and test circuit.
4/ As a minimum, gate to source leakages and drain to source leakage are to be examined to verify the electrical performance of the DUT prior to and after test. At the manufacturer's option, the remaining static tests in table IV, with the exception of step 8, may be performed.
5/ Devices passing a given combination of drain and gate voltage for an LET of 36 to 40 MeV-cm$^2$/mg qualify the same conditions for an LET of 26 to 30 MeV-cm$^2$/mg
6/ This sampling plan applies to each bias condition defined.



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TABLE IV. Group A, B, C and E electrical measurements

| Step | Inspection 1/, 2/, 3/ | MIL-STD-750 | | Symbol | Limits | | Units |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1. | Breakdown voltage drain to source | 3407 | $V_{GS}$ = 0V, $I_D$ = 1 mA dc, Bias condition C | $V_{(BR)DSS}$ | | | |
| | 2N7395 | | | | 100 | | V dc |
| | 2N7396 | | | | 200 | | V dc |
| | 2N7397 | | | | 250 | | V dc |
| | 2N7398 | | | | 500 | | V dc |
| 2. | Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1 mA dc | $V_{GS(TH)1}$ | 1.5 | 4.0 | V dc |
| 3. | Gate current | 3411 | $V_{GS}$ = +20V and -20V dc, Bias condition C, $V_{DS}$ = 0V | $I_{GSS1}$ | | ±100 | nA dc |
| 4. | Drain current | 3413 | $V_{GS}$ = 0V dc, Bias condition C, $V_{DS}$ = 80% of rated $V_{DS}$ | $I_{DSS1}$ | | 25 | μA dc |
| 5. | Static drain to source "ON"-state resistance | 3421 | $V_{GS}$ = 12V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D2}$ | $r_{DS(ON)1}$ | | | |
| | 2N7395 | | | | | 0.23 | Ω |
| | 2N7396 | | | | | 0.46 | Ω |
| | 2N7397 | | | | | 0.61 | Ω |
| | 2N7398 | | | | | 2.50 | Ω |
| 6. | Static drain to source "ON"-state voltage | 3405 | $V_{GS}$ = 12V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D1}$ | $V_{DS(ON)}$ | | | |
| | 2N7395 | | | | | 1.93 | V dc |
| | 2N7396 | | | | | 2.42 | V dc |
| | 2N7397 | | | | | 2.56 | V dc |
| | 2N7398 | | | | | 5.25 | V dc |
| 7. | Forward voltage | 4011 | $V_{GS}$ = 0 V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D1}$ | $V_{SD}$ | | 1.8 | V dc |
| 8. | Thermal response | 3161 | See 4.5.3 | $\Delta V_{SD}$ | | 125 | mV |

1/ The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:
a. Subgroup 3, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.
b. Subgroup 4, see table IV herein, steps 1, 2, 3, 4, 5, 6, 7, and 8.
c. Subgroup 5, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.

2/ The electrical measurements for table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows:
a. Subgroup 2, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.
b. Subgroup 3, see table IV herein, steps 1, 2, 3, 4, 5, 6, 7, and 8.

3/ The electrical measurements for table VII of MIL-PRF-19500 are as follows:
a. Subgroup 2 and 3, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.
b. Subgroup 6, see table IV herein, steps 1, 2, 3, 4, 5, 6, 7, and 8.



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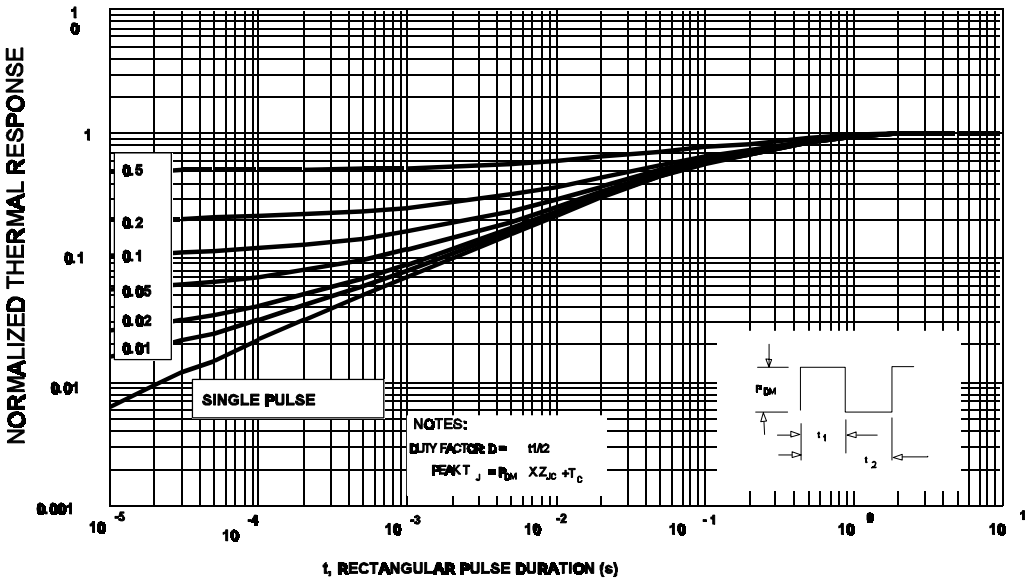



FIGURE 2. Thermal response curves.



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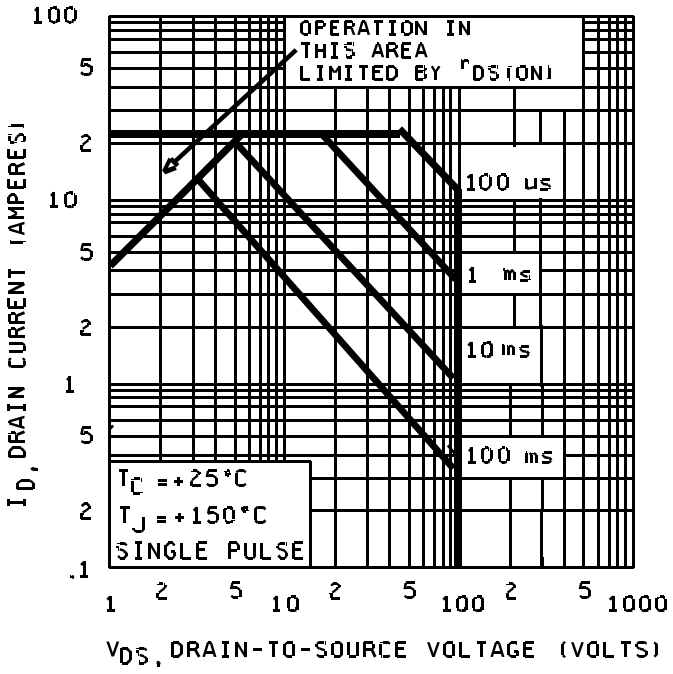



FIGURE 3. Safe operating area graphs.



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2N7396

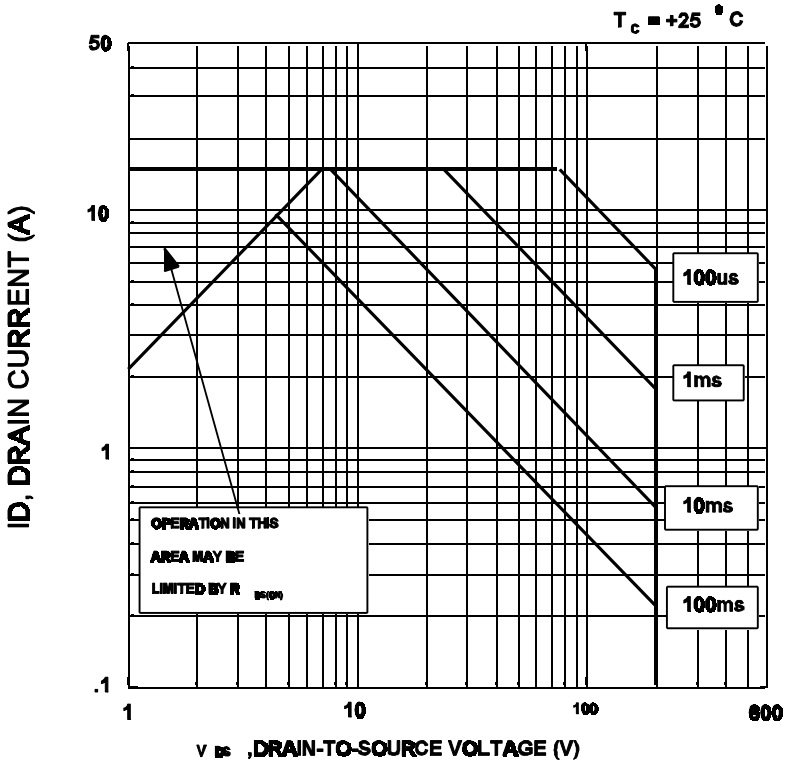



FIGURE 3. Safe operating area graphs - Continued



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2N7397

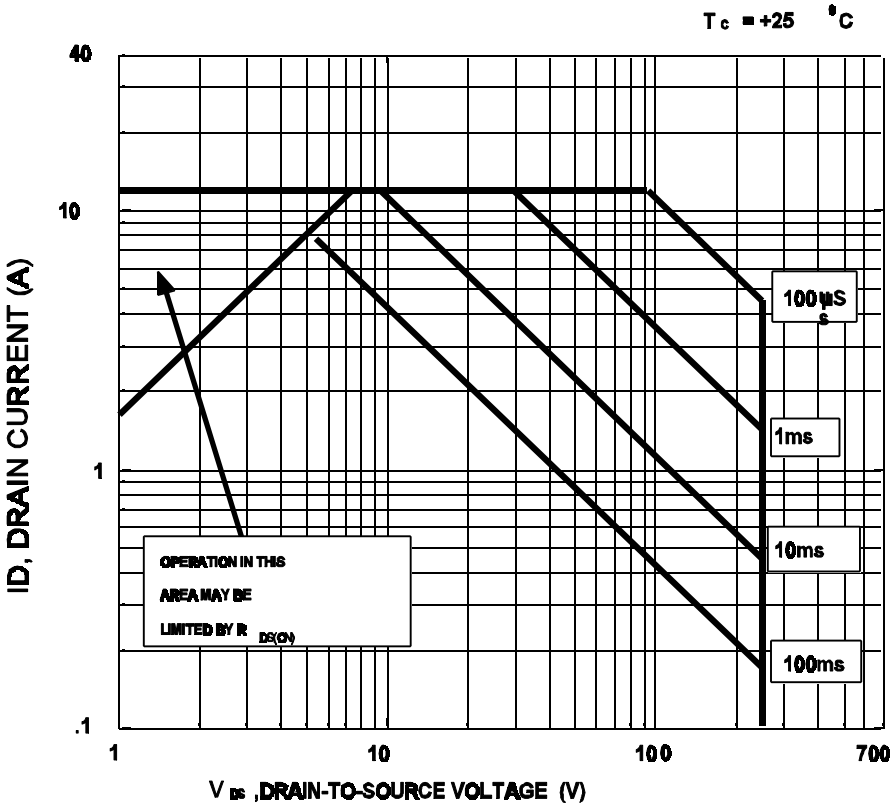



FIGURE 3. Safe operating area graphs - Continued



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2N7398

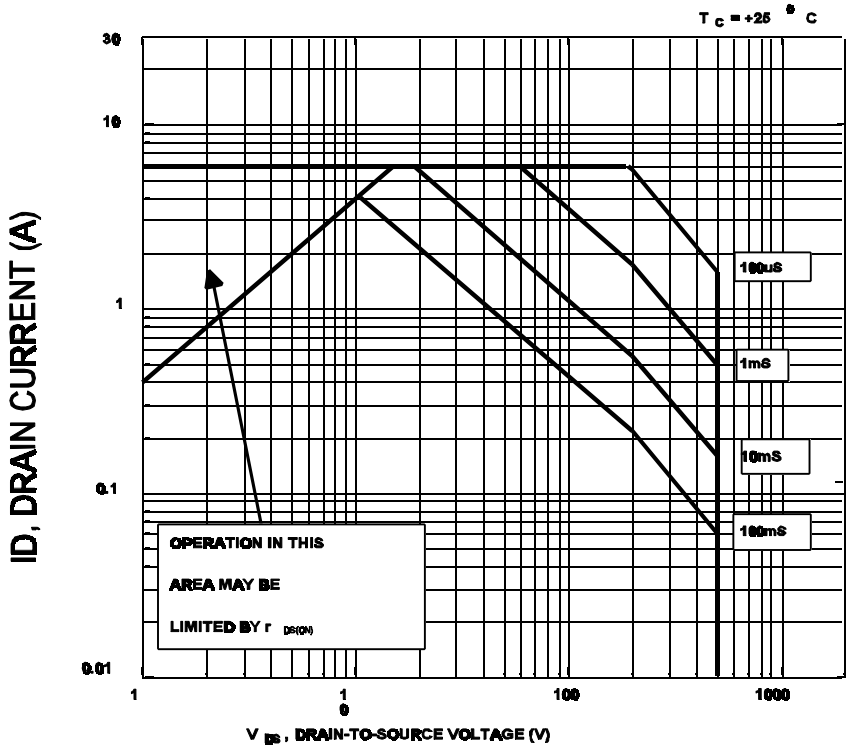



FIGURE 3. Safe operating area graphs - Continued.



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2N7395

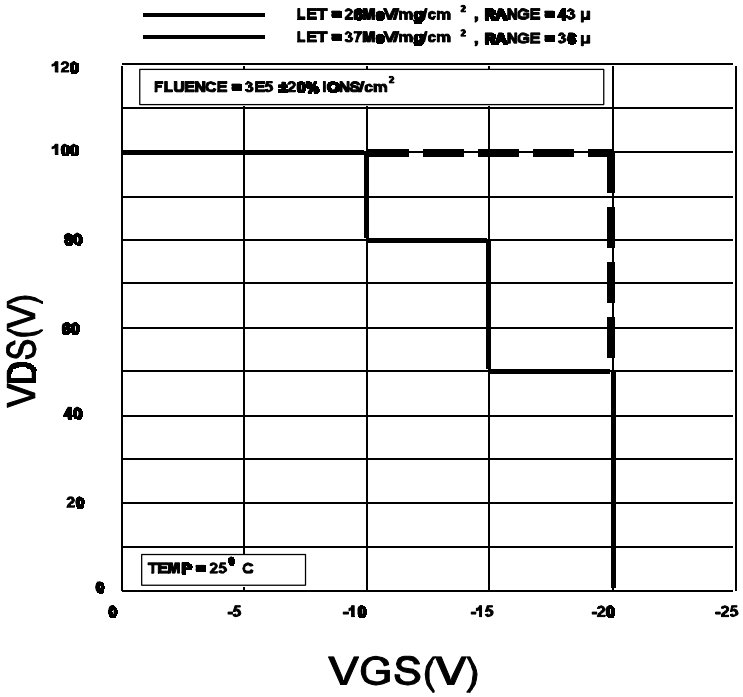



FIGURE 4. Single event effects safe operating area graphs.



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2N7396

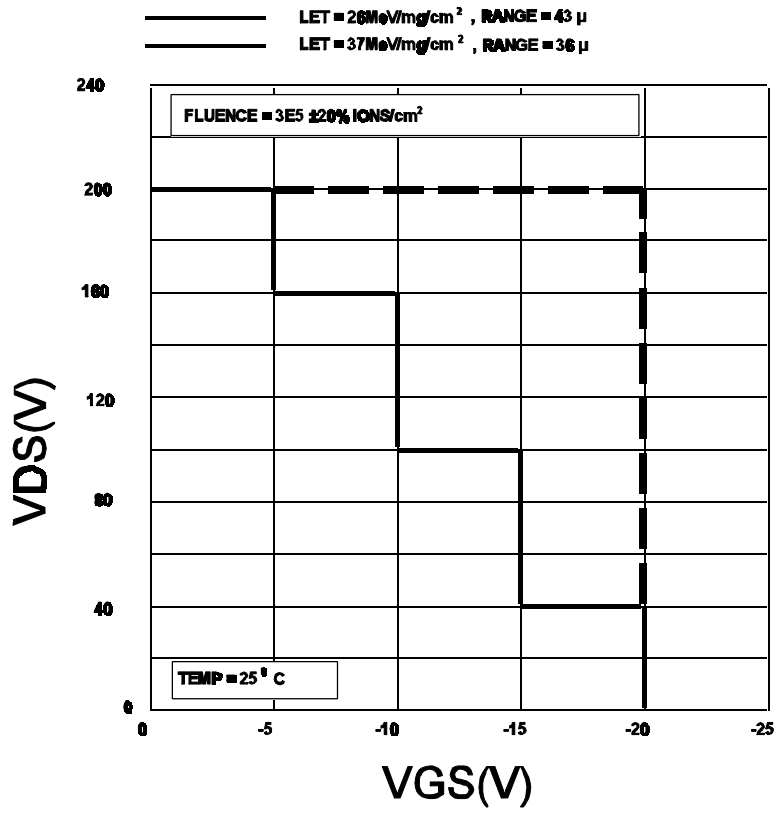



FIGURE 4. Single event effects safe operating area graphs - Continued.



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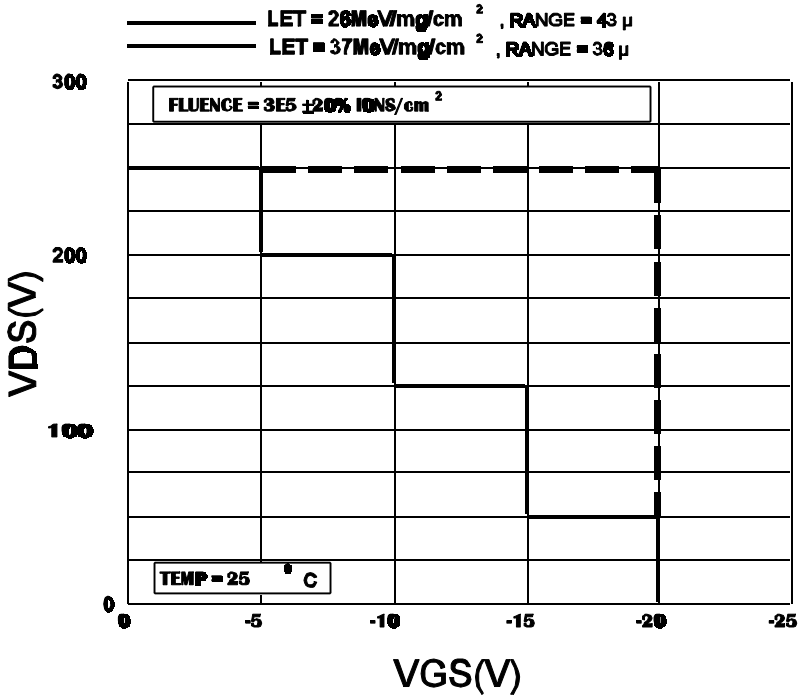



FIGURE 4. Single event effects safe operating area graphs - continued.



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2N7398

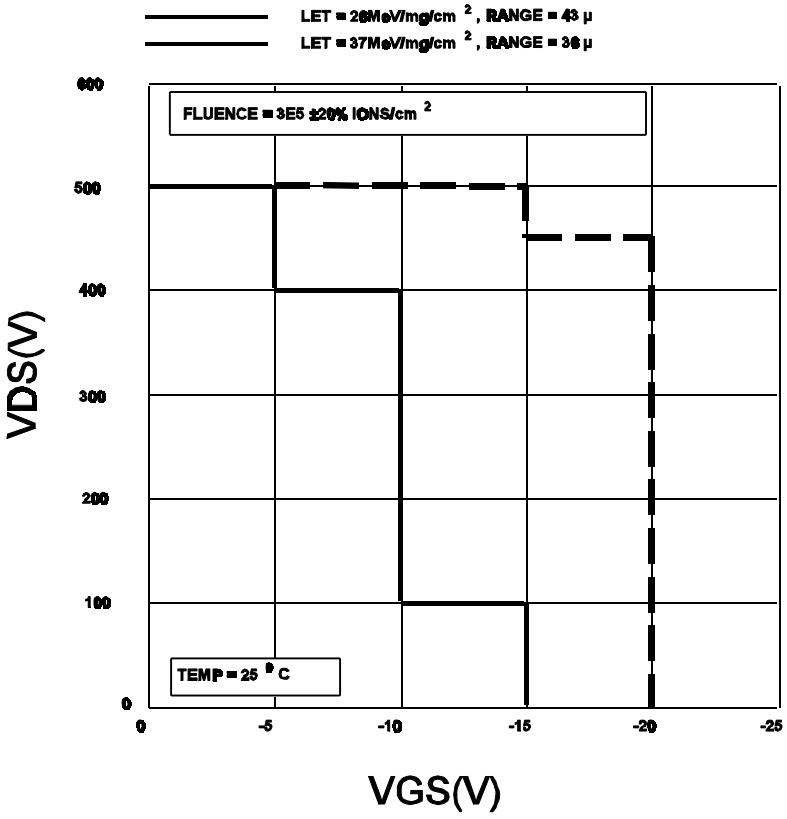



FIGURE 4. Single event effects safe operating area graphs - Continued.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of specification.

b. Issue of DODISS to be cited in the solicitation., title, number and date of specification.

c. Lead finish or formation may be specified (see 3.3.1).

d. Type designation and product assurance level.

6.3 Cross-reference list. The following table shows the generic P/N and its associated military P/N (without JAN and RHA prefix).

| Generic P/N | Military P/N |
|---|---|
| SL130 | 2N7395 |
| SL230 | 2N7396 |
| SL234 | 2N7397 |
| SL430 | 2N7398 |

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.



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APPENDIX A

1 Single Event Gate Rupture and Drain Burnout Tests

1.1 Scope. This test method defines the requirements for qualification testing of discrete power FETs, for single-event gate rupture (SEGR) and single-event drain burnout (SEB) resulting from energetic heavy ion irradiation from an accelerator. This procedure can be used with either steady state or pulse type irradiations. This test is destructive, it requires delidding of devices for most sources and is likely to result in device degradation or destruction.

1.2 SEGR/SEB Screening. All testing requires a chosen drain to source capacitor sufficiently large to damage the device in the event that SEB occurs. Accordingly, there must be no significant limiting drain or source resistance as is commonly used to obtain an SEB "error count" for past practices. Accordingly, SEB becomes a destructive test just as it is in a normal application. Although there will sometimes be confusion about the nature of the failure, (SEGR or SEB), it is moot.

1.2.2 Assurance testing. Assurance testing requires a chosen sample size and number of acceptable failures. Assurance testing requires a fixing of $V_{DS}$, $V_{GS}$, LET, energy, fluence, an approximate flux, angle of incidence (zero, which is the most severe), and temperature (SEGR is independent of temperature and SEB is slightly more severe at low temperature).

1.3 Apparatus

1.3.1 Power Supply. Using the test circuit of figure 5, regulated DC power supply with floating output to produce the voltages required to bias the junction.

1.3.2 Oscilloscope. Either a single dual-beam, or two single-beam oscilloscopes that have adequate bandwidth capability to ensure that radiation response and peak steady-state values are accurately displayed.

1.3.3 Oscilloscope Camera(s) and Film. Oscilloscope Camera(s) and Film capable of recording single transient traces at a sweep rate consistent with good resolution at the pulse width used in the test.

1.3.4 Digitizers. Digitizers with Bandwidth, Sampling Interval and Time-base Capabilities adequate for handling the transient signals with good resolution for all pulse widths utilized in the test may be used. Hard copy printouts of the recorded signal may be a part of the capability of this apparatus.

1.3.5 Cabling. Cabling to complete the connection of the test circuit in the exposure area with the power supply and oscilloscopes in the data area. Any type of ungrounded wiring may be used to connect the power supply to the bias points of the test circuit; however, coaxial cables properly terminated at the oscilloscope input are required for the signal leads.

1.4 Sampling. This is a destructive test. Therefore, for the purposes of lot testing or lot qualification, a representative sample should be drawn from the lot in accordance with a test plan that is agreed upon by parties to test. Any suitable package may be used for this test.

1.5 SEGR Test Plan A test plan shall be developed which specifies the following:

a. Devices to be tested and description.
b. Sample selection process and number of samples.
c. Bias and ambient conditions.
d. Bias step increments.
e. Dosimetry.
f. Method of monitoring the DUTs.
g. Temperature monitoring.
h. Beam condition including ion species to be utilized in performing the tests.
i. Data recording requirements.
j. Preliminary analysis, if required to establish a theoretical maximum $V_{DS}$-LET.



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1.6 Preparation of Apparatus The following steps should be followed in the preparation of the apparatus:

a. Utilize the appropriate test circuit (see figure 5).

b. Perform the dosimetry calibration and record.

c. Verify the response of the test circuit and associated cables:

d. Connect the device tester to the DUT.

e. Adjust $V_{DS}$ to a value $V_{DS}$ = 0.

1.6.1 Beam Tuning. The beam shall be adjusted to provide the required uniformity, level of contamination, and energy.

1.6.2 DUT Preparation.

a. Verify the electrical performance of the DUT.

b. Delid the device.

c. Any organic overcoats must be removed before testing is initiated.

d. Reverification of the electrical performance of device after delidding.

1.7 Device Screening.

a. Mount the DUT in the test circuit and place the assembly into the vacuum chamber.

b. Measure and record the ambient temperature in the vicinity of the DUT.

c. Verify the beam tuning.

d. Apply bias to the DUT as defined in the test plan.

e. Irradiate the DUT with the selected heavy ion.

f. Observe the oscilloscope or equivalent output measurement to verify DUT's operation.

g. Reverification of the electrical performance of device after ion exposure.

1.8 Test Reporting. The test data sheet shall contain the following information:

a. Dates, times, names of test personnel.
b. Type of accelerator, name and location; ion species and energy.
c. DUT types, serial numbers, functional description, technology, manufacturer, date code and mask number if known.
d. Device duty factor and fractional portion of the chip tested, if applicable.
e. Purpose for each test run and any changes from previous test run.
f. DUT operating parameters (bias, clock frequency, temperature, etc)
g. DUT test patterns or operational modes, including duty factor.
h. Beam angle.
i. Particle count (related to fluence), run time.
j. Number of errors, locations, and special comments (anomalous incidents).
k. Transient events and recovery time when instrumental.
l. Special test results; e.g., SEB, SEGR, etc.



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1.8.1. The test report shall contain the following information:

a. Test objective

b. Complete description of the product(s) tested.

c. Description of test setup and methods.

d. Complete description of bias and ambient conditions.

e. Summary of results.

f. Conclusions.

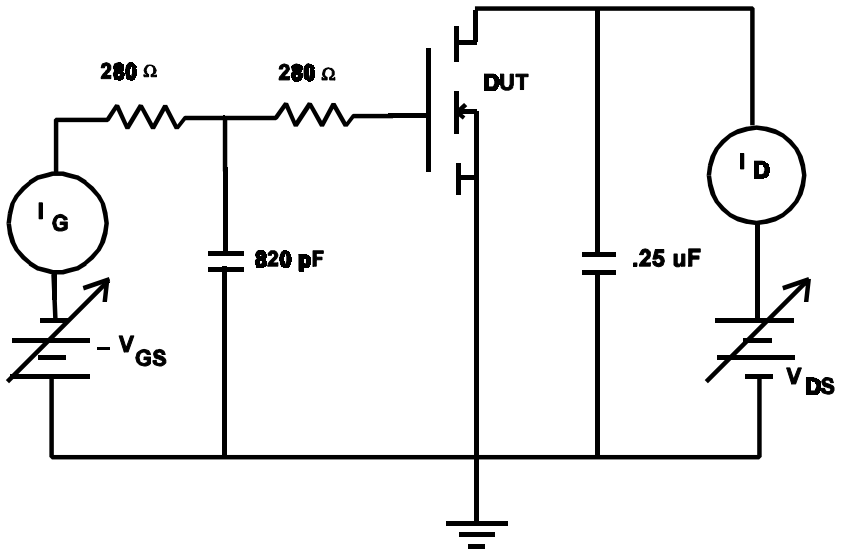



FIGURE 5. SEGR test circuit.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Preparing activity:
DLA - ES

(Project 5961-1843)



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**STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL**

**INSTRUCTIONS**

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| **I RECOMMEND A CHANGE:** | **1. DOCUMENT NUMBER** MIL-PRF-19500/631 | **2. DOCUMENT DATE (YYMMDD)** 96/11/29 |
|---|---|---|

**3. DOCUMENT TITLE:** SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED (TOTAL DOSE AND SINGLE EVENT EFFECTS) TRANSISTORS, N-CHANNEL SILICON TYPES 2N7395, 2N7396, 2N7397, AND 2N7398 JANTXD , -R, JANTXVD , -R, AND JANSD, -R

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME (Last, First, Middle initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br><br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED (YYMMDD) |

**8. PREPARING ACTIVITY**

| a. NAME<br>Alan Barone | b. TELEPHONE (Include Area Code)<br>(1) Commercial 614-692-0510<br>(2) AUTOVON 850-0510 |
|---|---|
| c. ADDRESS (Include Zip Code)<br>Defense Supply Center Columbus<br>ATTN: DSCC-VAT<br>Columbus, OH 43210-5000 | **IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:**<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403,<br>Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |



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